Output Form (OF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: *MILU* | ***Context:*** *North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is fully aligned. MILU evaluates models using accuracy as the primary metric [Q3, Q15, Q16, Q75], with log-likelihood scoring for non-API models [Q52–Q55] and structured JSON generative output for API-based models [Q56, Q57]. The output form is a single label/answer index, which directly matches the deployment's binary correct/incorrect MCQ scoring requirement. Per-subject, per-language, per-shot result tables [Q99–Q132] provide granular evidence for Hindi-specific performance. The author-acknowledged caveat that log-likelihood evaluation may yield different results than generation-based or chain-of-thought evaluation [Q85] is relevant if the deployment's runtime inference is generative — the user should verify that production inference matches the evaluation method used to characterise the chosen model.

ID	Evidence
Q52	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q53	“We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020).” (p.5)
Q54	“"Specifically, the log-likelihood of an answer (a) given the question (x), i.e., logP(a\” (p.5)
Q55	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q99	“Table 11: Detailed subject-wise evaluation for ABHINAND/TAMIL-LLAMA-7B-INSTRUCT-V0.2 on MILU across different languages.” (p.22)
Q100	“The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.22)
Q101	“Table 12: Detailed subject-wise evaluation for AI4BHARAT/AIRAVATA on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.23)
Q102	“Table 13: Detailed subject-wise evaluation for BHABHAAI/GAJENDRA-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.24)
Q103	“Table 14: Detailed subject-wise evaluation for COGNITIVE-LAB/AMBARI-7B-BASE-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.25)
Q104	“Table 15: Detailed subject-wise evaluation for COGNITIVE-LAB/AMBARI-7B-INSTRUCT-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.26)
Q105	“Table 16: Detailed subject-wise evaluation for GENVRADMIN/ARYABHATTA-GEMMAGENZ-VIKAS-MERGED on MILU across different languages.” (p.27)
Q106	“Table 17: Detailed subject-wise evaluation for MANISHIITG/OPEN-ADITI-V6-LLAMA3 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.28)
Q107	“Table 18: Detailed subject-wise evaluation for NICKMALHOTRA/PROJECTINDUS on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.29)
Q108	“Table 19: Detailed subject-wise evaluation for SARVAMAI/OPENHATHI-7B-HI-V0.1-BASE on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.30)
Q109	“Table 20: Detailed subject-wise evaluation for TENSOIC/KAN-LLAMA-7B-BASE on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.31)

Q110	“Table 21: Detailed subject-wise evaluation for ABHINAND/MALAYALAM-LLAMA-7B-INSTRUCT-V0.1 on MILU across different languages.” (p.32)
Q111	“Table 22: Detailed subject-wise evaluation for ABHINAND/TELUGU-LLAMA-7B-INSTRUCT-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.33)
Q112	“Table 23: Detailed subject-wise evaluation for SMALLSTEPAI/MISAL-7B-INSTRUCT-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.34)
Q113	“Table 24: Detailed subject-wise evaluation for SMALLSTEPAI/MISAL-7B-BASE-V0.1 on MILU across different languages.” (p.35)
Q114	“Table 25: Detailed subject-wise evaluation for TELUGU-LLM-LABS/TELUGU-LLAMA2-7B-V0-BASE on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.36)
Q115	“Table 26: Detailed subject-wise evaluation for GPT-4o on MILU across different languages. The results reported are for 0-shot experiments.” (p.37)
Q116	“Table 27: Detailed subject-wise evaluation for GPT-4O-MINI on MILU across different languages. The results reported are for 0-shot experiments.” (p.38)
Q117	“Table 28: Detailed subject-wise evaluation for GEMINI-1.5-PRO on MILU across different languages. The results reported are for 0-shot experiments.” (p.39)
Q118	“Table 29: Detailed subject-wise evaluation for GEMINI-1.5-FLASH on MILU across different languages. The results reported are for 0-shot experiments.” (p.40)
Q119	“Table 30: Detailed subject-wise evaluation for KRUTRIM-SPECTRE-V2 on MILU across different languages. The results reported are for 0-shot experiments.” (p.41)
Q120	“Table 31: Detailed subject-wise evaluation for SARVAMAI/SARVAM-1 on MILU across different languages. The results reported are for 5-shot experiments.” (p.42)
Q121	“Table 32: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.2-1B on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.43)
Q122	“Table 33: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.2-1B-INSTRUCT on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.43)
Q123	“Table 34: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-2B on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.44)
Q124	“Table 35: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-2B-IT on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.44)
Q125	“Table 38: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.2-3B-INSTRUCT on MILU across different languages.” (p.46)
Q126	“Table 39: Detailed subject-wise evaluation for NVIDIA/NEMOTRON-4-MINI-HINDI-4B-BASE on MILU across different languages. The results reported are for 5-shot experiments.” (p.47)
Q127	“Table 42: Detailed subject-wise evaluation for NEULAB/PANGEA-7B on MILU across different languages. The results reported are for 5-shot experiments.” (p.49)
Q128	“Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.52)
Q129	“Table 48: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-27B on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively.” (p.52)
Q130	“Table 52: 0-shot subject-wise evaluation for META-LLAMA/LLAMA-3.1-70B-INSTRUCT on MILU across different languages.” (p.55)
Q131	“Table 53: 1-shot subject-wise evaluation for META-LLAMA/LLAMA-3.1-70B-INSTRUCT on MILU across different languages.” (p.56)
Q132	“Table 54: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.1-405B on MILU across different languages. The results reported are for 0-shot experiments.” (p.57)
Q3	“We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%.” (p.1)
Q15	“We evaluate 45 different LLMs - a mix of closed proprietary, open-source, and language-specific models- on MILU.” (p.2)
Q16	“Our findings suggest that models struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%.” (p.2)

Q75	“We evaluate 45 different LLMs and find that the majority of LLMs struggle on MILU, with GPT4o achieving the highest average accuracy.” (p.8)
Q56	“The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities.” (p.5)
Q57	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)

Strengths

- Accuracy metric directly matches binary MCQ correct/incorrect scoring [Q3, Q16]
- Both log-likelihood [Q52–Q55] and structured JSON generative [Q56, Q57] evaluation modes available
- Granular per-subject per-language reporting at 0/1/5-shot [Q99–Q132] enables Hindi-specific performance lookup
- Authors document the log-likelihood vs. generation caveat explicitly [Q85]

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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality is a single discrete answer choice, matching deployment requirement exactly [Q3, Q53–Q55]. No mismatch.

OF-2: Text-to-speech is not required for this deployment (text-only Hindi MCQ); no concern.

OF-3: Target population is postgraduate-qualified Hindi-medium teachers — high literacy assumed; no accessibility-driven output-form constraint identified.

OF-4: Documented caveat: log-likelihood evaluation may diverge from generation-based or CoT performance [Q85]. If deployment uses generative inference, model selection should be informed by API-mode evaluation rather than non-API log-likelihood numbers.

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Remediation

Gap 1: Log-likelihood evaluation may diverge from generative inference (acknowledged by authors)

Recommendation: If production deployment uses generative inference, base model selection on the API-mode (generative) MILU results rather than log-likelihood numbers, or run an internal generative re-evaluation on the candidate model